

AHMED GOMAA EL-SAYED

Data Analyst

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PROFESSIONAL SUMMARY

Detail-oriented Data Analyst with hands-on portfolio experience turning complex datasets into actionable business insights. Proficient in Excel, SQL, Python, and Power BI, with a strong focus on Customer Behavior Analysis, HR Workforce Analytics, Retail Omnichannel Audit, and Business Intelligence. Skilled in Data Cleaning, Data Validation, Data Storytelling, Anomaly Detection, ETL Pipeline Architecture, KPI Reporting, and Analytical Thinking to drive executive decision-making. Core competencies in data accuracy, recordkeeping, and reconciliation are directly transferable to inventory and stock reconciliation, sales and business reporting, and general office/administrative operations. Academic background in psychology provides a unique edge in interpreting behavioral and operational patterns, customer friction, and communicating clear, impactful findings to stakeholders. Completed military service and immediately available for full-time roles.

CORE COMPETENCIES

- Data Analysis & ETL: Data Accuracy & Validation, ETL Pipeline Architecture, Anomaly Detection & Anomaly Rules, Data Cleaning, Identity Resolution
- Reporting & Modeling: Business Intelligence, KPI Tracking, Star Schema Modeling, Dashboard Design, Business & Sales Reporting
- Business Acumen: Customer Behavior & Behavioral Analytics, Margin Leakage Detection, Inventory & Stock Reconciliation, Record-Keeping, Stakeholder Communication

TECHNICAL SKILLS

- Data Analysis: Microsoft Excel, Power Query, Pivot Tables, Power Pivot, Dashboard Design, Data Cleaning, Data Validation, Data Entry & Documentation
- SQL: PostgreSQL, MySQL, CTEs, Window Functions (ROW_NUMBER, LAG, LEAD), Joins, CASE WHEN, Aggregations, ETL Staging
- Python: Pandas, NumPy, Matplotlib, Seaborn, Isolation Forest, IQR Anomaly Detection, Exploratory Data Analysis (EDA)
- Business Intelligence: Excel Dashboards, KPI Reporting, Business Reporting, Power BI, DAX, Star Schema Data Modeling, Data Quality SLAs

PRACTICAL DATA ANALYTICS EXPERIENCE

Enterprise Omnichannel Audit & Margin Leakage Detection | 2026

Tools: SQL (PostgreSQL, CTEs, Window Functions), Python (Pandas, Isolation Forest, IQR), Power BI, DAX, Star Schema | Link: github.com/Ahmed1870/enterprise-omnichannel-audit

- Executed a simulated enterprise retail audit across Store, Web, Mobile App, and Marketplace channels on 108,000 transactions (\$6.98M Gross Sales) to identify margin erosion and revenue leakage.
- Engineered an end-to-end data pipeline and ETL workflow: automated CSV profiling, SQL staging, timestamp normalization, customer identity resolution, category mapping, and multi-currency (FX) conversion.
- Built a Star Schema dimensional model (1 Fact and 6 Dimension tables) to power Power BI semantic modeling and executive reporting.
- Applied advanced SQL (CTEs and Window Functions) to analyze cohort retention, return-risk segmentation, and policy compliance (>180-day window).
- Performed Python-based anomaly detection (IQR & Isolation Forest) to pinpoint promotional abuse (\$132.6K leakage proxy, 1.5% abuse rate) and suspicious returns (\$125.7K proxy, 44% risk rate).
- Leveraged psychological insights (wardrobing behavior, cognitive friction) to design selective friction rules and auditable controls, protecting revenue without harming customer retention.

- Developed interactive Power BI dashboards (Executive, Fraud Risk, Returns) with custom DAX measures tracking Net Sales (\$6.27M), Realized Margin (\$2.46M), and a Return Friction Index to drive business governance.

Customer Behavior & RFM Analysis | 2026

Tools: SQL, Python (Pandas, NumPy, Matplotlib), Excel, Power Query, Power BI | Link: github.com/Ahmed1870/Customer-Behavior-RFM-Analysis

- Cleaned and processed over 540,000 retail transactions (Online Retail Dataset) to ensure data integrity and establish a reliable foundation for customer-level analysis.
- Calculated RFM metrics and engineered customer-level analytical tables using advanced SQL (CTEs, Window Functions, and CASE WHEN logic).
- Segmented customers into actionable cohorts (e.g., Champions, At-Risk, Lost) and developed tailored retention strategies, identifying a potential recovery of up to 15% in at-risk revenue.
- Conducted Python-based EDA (Pandas, Matplotlib) to uncover purchasing patterns and built an interactive Excel dashboard to dynamically track core business KPIs.

Employee Attrition Analysis (HR Analytics) | 2026

Tools: SQL, Python (Pandas, NumPy), Power BI, DAX | Link: github.com/Ahmed1870/HR-Attrition-Data-Analysis

- Analyzed a 5,000-employee HR dataset (15% attrition rate) to uncover turnover drivers, utilizing SQL for data cleaning and Python for feature engineering (income quartiles, commute-distance bands, overtime flags).
- Built an interactive Power BI dashboard with custom DAX measures to visualize attrition trends across departments, roles, and demographic factors.
- Identified overtime as the primary attrition driver (29.3% vs. 8.7% turnover, a 3.4x gap) and flagged Executive and Analyst roles as unexpected high-risk segments.
- Delivered targeted retention strategies, proposing overtime caps for high-risk departments and compensation reviews for the lowest income quartile.

EDUCATION

Bachelor of Arts – Psychology — Graduated: 2024

Faculty of Arts, Fayoum University, Egypt

Relevant Coursework: Research Methods, Statistical Analysis, Behavioral Data Interpretation, Academic Report Writing

CERTIFICATIONS & TRAINING

- Google Data Analytics Professional Certificate — Google via Coursera (2024)
- Excel Skills for Business — Macquarie University via Coursera (2024)
- SQL for Data Science — UC Davis via Coursera (2024)
- Data Analysis with Python — IBM via Coursera (2024)
- Microsoft Power BI Desktop for Business Intelligence — Udemy (2024)

ADDITIONAL INFORMATION

- Languages: Arabic (Native), English (Professional Working Proficiency)
- Military Service: Completed – No obligations